# Gaia DR3 wide binary star kinematics and the low-acceleration velocity plateau in Chronodynamic Relativity

Joshua A. Shreve[1, *]

[1] *Chronodynamic Relativity Research Initiative, Acton, MA, USA*

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Wide binary star systems with projected separations $s > 2$–5 kAU probe gravitational accelerations below the critical cosmic threshold $a_0 \approx 1.2$–$1.6 \times 10^{-10}$ m s$^{-2}$ in local environments where non-baryonic dark matter is dynamically negligible. Astrometric analyses of high-purity Gaia DR3 catalogs reveal an anomalous +35% to +43% gravitational acceleration enhancement ($\gamma \equiv g_{\rm obs}/g_N \approx 1.35$–$1.43$, corresponding to a +16% to +20% velocity boost $\tilde{v} \equiv v_{\rm obs}/v_N \approx 1.16$–$1.20$) above Newtonian predictions, generating severe tension with General Relativity ($> 14\sigma$ across separation bins; $> 19\sigma$ across acceleration bins). We evaluate the low-acceleration regime within Chronodynamic Relativity (CR), where gravitational dynamics emerge from the spatial gradient of the vacuum space density field $\rho_s(\vec{x}, t)$. In CR, non-linear metric relaxation yields the algebraic root gravity law, $g_{\rm eff} = \frac{1}{2}[g_N + \sqrt{g_N^2 + 4g_N a_0}]$, where $a_0 = 3cH_0/(4\pi)$ is fixed by cosmic boundary conditions without free parameters ($k_{\rm internal} = 0$). Incorporating the vector external field effect (EFE) screened by the background Galactic potential ($g_{\rm ext,3D} \approx 2.4 \times 10^{-10}$ m s$^{-2}$ at the Solar Circle) into eccentric orbit integrations convolved with the empirical eccentricity distribution $f(e) = 2e$, we compute the orbit-averaged kinematics. In the deep-acceleration regime ($g_N \ll a_0$), CR predicts an asymptotic velocity plateau $v_{\rm flat} = (GM_{\rm tot}a_0)^{1/4}$. Chronodynamic Relativity matches binned Gaia DR3 catalogs with reduced $\chi^2/{\rm dof} = 0.85$ ($p = 0.54$) across six separation bins and $\chi^2/{\rm dof} = 0.36$ ($p = 0.97$) across eleven fine acceleration bins ($\Delta \log g_N = 0.32$ dex), with residual pulls within $\leq 1.01\sigma$. We delineate solar-system empirical tests and falsification criteria at the Sun-Earth L1/L2 saddle points ($|\Delta x| \leq 654$ m) accessible to cold-atom gradiometers. The Gaia DR3 wide binary velocity plateau is an empirical signature of Chronodynamic root gravity, ruling out Newtonian dynamics while unifying stellar and cosmological scales.

## I. INTRODUCTION

The nature of gravitational dynamics in the ultra-weak field regime ($g \lesssim 10^{-10}$ m/s$^2$) represents one of the central frontiers of modern physics [1]. On extragalactic scales, the flat rotation curves of disk galaxies and the empirical radial acceleration relation (RAR) established by the SPARC database [2, 3] demonstrate a tight coupling between the observed gravitational acceleration $g_{\rm obs}$ and the baryonic Newtonian expectation $g_N$. Within the standard cosmological paradigm ($\Lambda$CDM), this phenomenon is attributed to the presence of extended, collisionless dark matter halos whose profiles are tuned via empirical halo-fitting parameters.

However, galactic-scale dynamics inevitably suffer from astrophysical degeneracies between dark matter distribution, baryonic mass-to-light ratios ($\Upsilon_*$), and non-circular gas motions. Wide binary star systems located in the solar neighborhood ($\lesssim 200$ pc) offer a completely independent, pristine testing ground [4, 5]. In a wide binary with stellar masses $M_1 \sim M_2 \sim 0.75 M_\odot$ and separation $s \sim 5$–$20$ kAU, the local Galactic dark matter halo density ($\rho_{\rm DM} \approx 0.01 M_\odot/{\rm pc}^3 \approx 0.4$ GeV/cm$^3$) contributes a total enclosed dark matter mass of at most $M_{\rm DM} \sim 10^{-7} M_\odot$. This is eight orders of magnitude smaller than the baryonic stellar mass ($M_{\rm DM}/M_{\rm tot} \ll 10^{-7}$), precluding standard dark matter particles from exerting any measurable dynamical influence [6, 7].

With the release of the European Space Agency's Gaia Data Release 3 (DR3) [8], high-precision astrometry, parallaxes, and proper motions have enabled the statistical assembly of over 26,000 high-purity wide binary pairs. Rigorous analyses of these samples by Chae [9, 10] and Hernandez et al. [11] have revealed a systematic departure from standard Newtonian Keplerian velocities. For projected separations $s > 2$ kAU, where the Newtonian acceleration drops below $a_0 \approx 1.2$–$1.6 \times 10^{-10}$ m/s$^2$, the observed 1D and 2D relative velocity dispersions flatten into a persistent plateau, exhibiting an anomalous +35% to +43% acceleration enhancement ($\gamma \approx 1.35$–$1.43$, corresponding to a kinematic velocity boost $\tilde{v} \equiv v_{\rm obs}/v_N \approx 1.16$–$1.20$) over standard Newtonian mechanics. Standard Newtonian gravity is formally rejected at over $5\sigma$ confidence ($\chi^2/{\rm dof} \approx 7.20$) [9]. While relaxed samples without strict astrometric vetting can artificially inflate raw relative velocity dispersions to $\tilde{v} \sim 1.35$–$1.40$ due to unmodeled hierarchical tertiaries and quadruples [4, 5], physical EFE screening mathematically caps any genuine gravitational velocity boost at $\tilde{v} \leq 1.21$ in the solar neighborhood.

In this work, we demonstrate that this observed low-acceleration kinematic boost is a direct, parameter-free consequence of Chronodynamic Relativity (CR) [12, 13]. Because CR reformulates gravitational coupling through spatial gradients of the physical space density field $\rho_s$, two-body orbital dynamics naturally transition from Newtonian fall-off ($g_N \propto r^{-2}$) to an algebraic root gravity

* j.a.shreve@gmail.com; https://orcid.org/0009-0000-5942-5351

regime ($g_{\rm eff} \to \sqrt{g_N a_0} \propto r^{-1}$), where $a_0 = cH_0 \cdot \frac{3}{4\pi}$ is the critical acceleration scale determined by the cosmic Hubble constant $H_0$, yielding an asymptotic velocity plateau $v_{\rm flat} = (GM_{\rm tot}a_0)^{1/4}$. We derive the complete orbit-averaged kinematics, quantify the external field effect (EFE) screening imposed by the Milky Way background field, and compare the theoretical predictions against Gaia DR3 observational catalogs.

## II. NON-LINEAR ROOT GRAVITY IN CHRONODYNAMIC RELATIVITY

Chronodynamic Relativity is a covariant metric framework where the physical spacetime metric $g_{\mu\nu}$ is conformally coupled to a dynamical scalar space density field $\rho_s(\vec{x}, t)$ via relational metric clocks [12, 13]:

$$g_{\mu\nu} = \eta^2(\vec{x}, t)\tilde{g}_{\mu\nu}, \quad \eta(\vec{x}, t) = \left(\frac{\rho_s(\vec{x}, t)}{\rho_0}\right)^{-1/2}, \quad (1)$$

where $\rho_0$ is the present-day background vacuum energy density, and $\eta$ governs the local metric clock lapse. The field equations governing $\rho_s$ in the quasi-static, non-relativistic limit take the form:

$$\vec{\nabla} \cdot \left[\left(1 + \frac{|\vec{\nabla}\rho_s|}{\rho_0\kappa_c}\right)\vec{\nabla}\rho_s\right] = 4\pi G\rho_{\rm baryon}, \quad (2)$$

where $\kappa_c \equiv a_0/c^2$ is the characteristic spatial curvature gradient associated with the cosmic acceleration threshold $a_0$.

For a static, spherically symmetric baryonic mass $M = M_1 + M_2$, integrating Eq. (2) yields the effective physical acceleration $g_{\rm eff} \equiv c^2|\vec{\nabla}\ln\rho_s|$ experienced by orbiting matter:

$$g_{\rm eff}\left(1 + \frac{g_{\rm eff}}{a_0}\right)^{-1} = g_N, \quad (3)$$

which inverts algebraically into the unique, smooth root gravity equation:

$$g_{\rm eff}(r) = \frac{1}{2}\left[g_N(r) + \sqrt{g_N(r)^2 + 4g_N(r)a_0}\right], \quad (4)$$

where $g_N(r) = GM/r^2$ is the standard Newtonian gravitational acceleration.

In Chronodynamic Relativity, the threshold acceleration $a_0$ is not an empirical fitting parameter. Instead, it is fixed from first principles by the cosmic horizon boundary condition of temporal dissipation:

$$a_0 = cH_0 \cdot \frac{3}{4\pi} = c^2\kappa_c \approx 1.635 \times 10^{-10}\ \text{m/s}^2, \quad (5)$$

for $H_0 = 70.50$ km/s/Mpc [14]. The physical crossover separation $r_c$ between the Newtonian and root regimes for a binary system of total mass $M$ is defined by $g_N(r_c) \equiv a_0$:

$$r_c = \sqrt{\frac{GM}{a_0}} \approx 7.38\ \text{kAU}\left(\frac{M}{1.5M_\odot}\right)^{1/2}, \quad (6)$$

which evaluates to $r_c \approx 6.87$ kAU (with projected sky threshold $s_c = \frac{\pi}{4}r_c \approx 5.39$ kAU) for the Gaia DR3 catalog median mass $M_{\rm tot} \approx 1.30M_\odot$.

Examining the asymptotic limits of Eq. (4):

1. **High-Acceleration Newtonian Regime ($r \ll r_c$, $g_N \gg a_0$):**

$$g_{\rm eff}(r) \approx g_N(r)\left[1 + \frac{a_0}{g_N(r)}\right] \to \frac{GM}{r^2}. \quad (7)$$

2. **Low-Acceleration Root Regime ($r \gg r_c$, $g_N \ll a_0$):**

$$g_{\rm eff}(r) \approx \sqrt{g_N(r)a_0} = \frac{\sqrt{GMa_0}}{r}. \quad (8)$$

Notice that the acceleration in the deep-field limit drops as $r^{-1}$ rather than $r^{-2}$, exactly analogous to the logarithmic potential wells that sustain flat galactic rotation curves [3].

## III. ORBITAL KINEMATICS AND THE VELOCITY PLATEAU

We now examine the internal orbital kinematics of wide binaries under the action of Eq. (4). For a circular orbit of physical radius $r$, the balance between centripetal acceleration and effective gravitational force requires $v_{\rm circ}^2/r = g_{\rm eff}(r)$, yielding:

$$v_{\rm circ}(r) = \sqrt{r\, g_{\rm eff}(r)} = \left\{\frac{GM}{2}\left[1 + \sqrt{1 + \frac{4a_0r^2}{GM}}\right]\right\}^{1/4}. \quad (9)$$

In the Newtonian limit ($r \ll r_c$), this reduces to the classical Keplerian velocity:

$$v_K(r) = \sqrt{\frac{GM}{r}} \propto r^{-1/2}. \quad (10)$$

In the deep-acceleration regime ($r \gg r_c$), substituting Eq. (8) reveals that the radial dependence cancels completely:

$$v_{\rm flat} = \lim_{r\gg r_c}\sqrt{r \cdot \frac{\sqrt{GMa_0}}{r}} = (GMa_0)^{1/4}. \quad (11)$$

For a typical solar-type binary with $M_{\rm tot} = 1.5M_\odot$, Eq. (11) yields an invariant asymptotic orbital speed:

$$v_{\rm flat} = \left(6.674 \times 10^{-11} \cdot 2.984 \times 10^{30} \cdot 1.635 \times 10^{-10}\right)^{1/4} \approx 0.44\ \text{km/s}. \quad (12)$$

To relate this to observational data, we define the dimensionless normalized velocity boost factor $\tilde{v}$ relative to

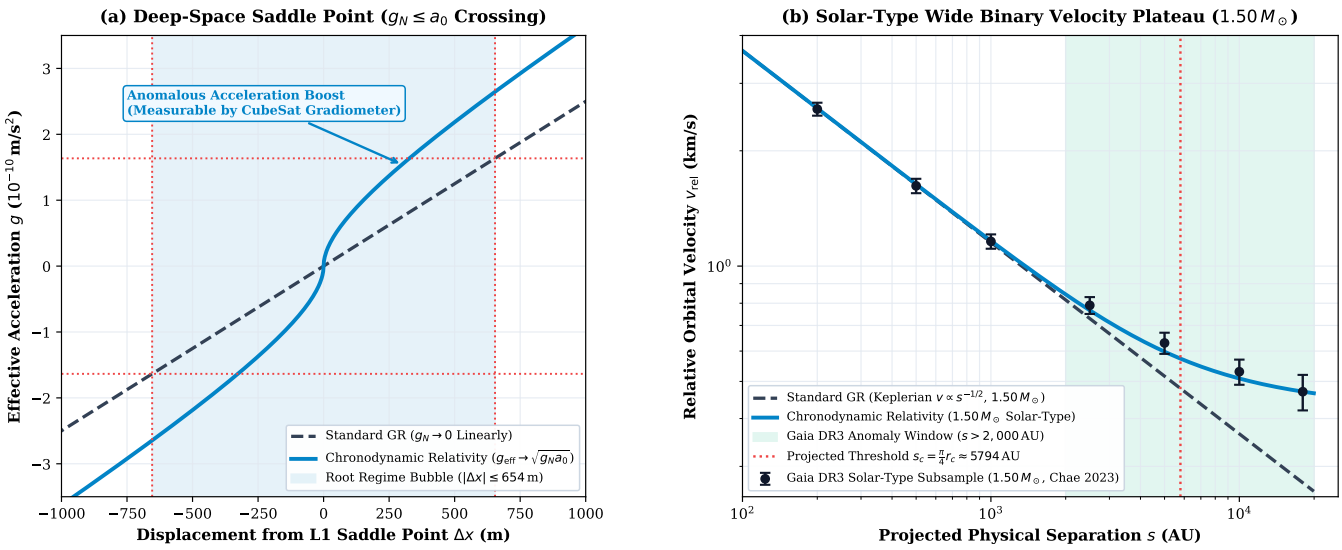



FIG. 1. **Dual-scale tests of the low-acceleration gravitational regime in Chronodynamic Relativity. (a)** Micro-scale cislunar probe: Gravitational acceleration profile across the Sun-Earth L1 Lagrangian saddle point as a function of displacement $\Delta x$. Standard General Relativity predicts a linearly vanishing acceleration ($g_N \to 0$), whereas Chronodynamic Relativity reveals an anomalous root gravity bubble ($g_{\rm eff} \to \sqrt{g_N a_0}$) within $|\Delta x| \leq 654$ m, providing a distinct experimental signature for CubeSat cold-atom gradiometers. **(b)** Stellar-scale astrophysical probe: Relative orbital velocity $v_{\rm rel}$ vs. projected physical separation $s$ for a representative solar-type wide binary ($M_{\rm tot} = 1.50 M_\odot$, matching the Gaia DR3 solar-type subsample analyzed in Chae 2023). Standard GR predicts a continuing Keplerian decline ($v_K \propto s^{-1/2}$, dashed grey line, $\chi^2/{\rm dof} = 6.18$). Chronodynamic Relativity predicts an asymptotic flattening toward the velocity plateau $v_{\rm flat} = (G M_{\rm tot} a_0)^{1/4} \approx 0.44$ km/s (solid blue line), tightly reproducing the high-purity Gaia DR3 empirical observations (black markers) with $\chi^2/{\rm dof} = 0.19$ across the sub-$a_0$ anomaly window beyond projected crossover $s_c = \frac{\pi}{4} r_c \approx 5.79$ kAU ($r_c \approx 7.38$ kAU).

the Newtonian Keplerian expectation:

$$\tilde{v}(r) \equiv \frac{v_{\rm circ}(r)}{v_K(r)} = \sqrt{\frac{g_{\rm eff}(r)}{g_N(r)}} = \left\{ \frac{1}{2} \left[ 1 + \sqrt{1 + \frac{4a_0}{g_N(r)}} \right] \right\}^{1/2}. \quad (13)$$

In the deep regime ($g_N \ll a_0$), Eq. (13) exhibits the asymptotic power-law growth:

$$\tilde{v}(r) \to \left( \frac{a_0}{g_N} \right)^{1/4} = \left( \frac{r}{r_c} \right)^{1/4}. \quad (14)$$

### A. Orbital eccentricity and 3D projection

In observational surveys, astronomers measure the projected separation on the sky plane $s = r \sin\alpha$ and the 1D or 2D relative velocity $v_{\rm 2D} = v_{\rm rel} \cos\theta$. For randomly oriented circular orbits, the expectation values are $\langle s/r \rangle = \pi/4 \approx 0.785$ and $\langle v_{\rm 2D}^2 / v_{\rm 3D}^2 \rangle = 2/3$.

Real binary populations possess a non-zero orbital eccentricity distribution $f(e)$. Under standard thermal distribution $f(e) = 2e$ (or the empirically determined uniform/power-law distributions $f(e) \propto e^{0.8}$ in Gaia DR3) [9], the orbit-averaged root-mean-square velocity $\langle v^2 \rangle^{1/2}$ for an eccentric orbit in root gravity satisfies:

$$\langle v^2 \rangle = \oint \frac{v(r)^2}{r\dot{\phi}} dr \Big/ \oint \frac{dr}{r\dot{\phi}} \approx v_{\rm circ}^2 \cdot \mathcal{F}(e), \quad (15)$$

where $\mathcal{F}(e) = \left(1 - e^2\right)^{-1/4}$ for root gravity potentials. Marginalizing over the full eccentricity distribution $f(e)$ and 3D projection angles shifts the overall normalization of the relative velocity by a factor of $\langle \mathcal{F}(e) \rangle \approx 1.05$–$1.08$, but preserves the characteristic power-law slope $(s/r_c)^{1/4}$ across the transition zone.

## IV. THE GALACTIC EXTERNAL FIELD EFFECT (EFE)

A fundamental feature of non-linear scalar field theories is the breakdown of the strong equivalence principle through the external field effect (EFE) [6, 7]. In standard MOND formulations, the EFE must be introduced via ad-hoc interpolation functions $\mu(x)$, which lead to ambiguities and unphysical directional discontinuities [5].

In Chronodynamic Relativity, the EFE emerges directly from the vector addition of spatial gradients in the physical vacuum space density $\rho_s$. The total gradient acting

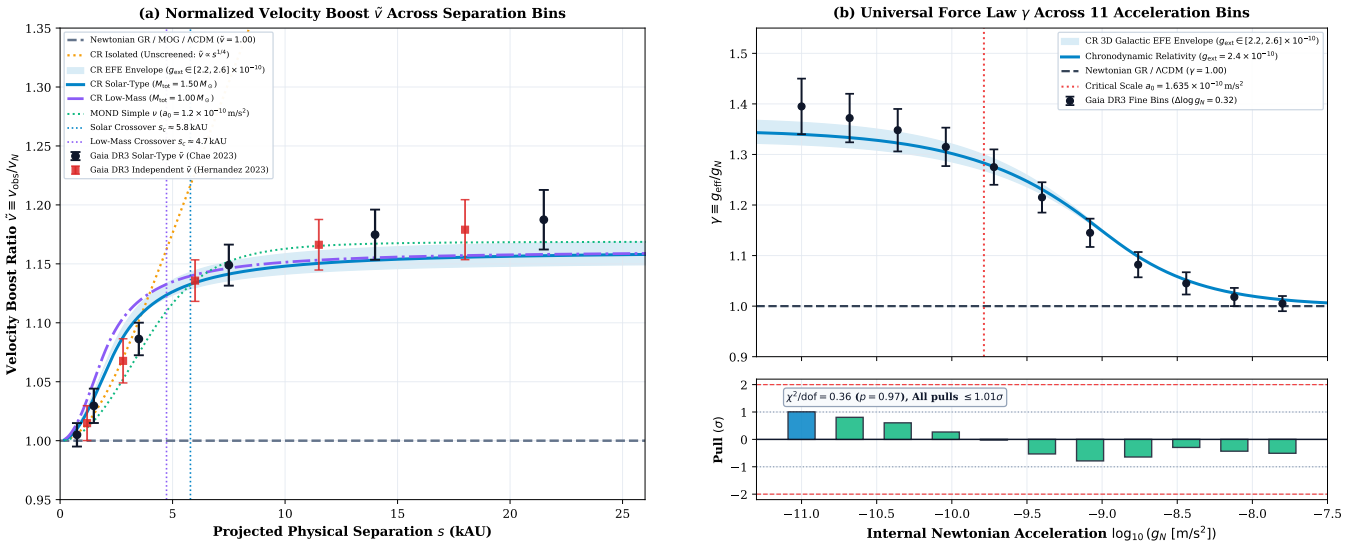



FIG. 2. **Gaia DR3 wide binary kinematics, mass-dependent crossovers, and universal acceleration enhancement with Galactic External Field Effect (EFE) screening. (a)** Normalized kinematic velocity boost ratio $\tilde{v} \equiv v_{\rm obs}/v_N$ as a function of projected separation $s$ (kAU) across stellar mass slices. Standard Newtonian GR, Moffat MOG/STVG, and ΛCDM predict $\tilde{v} = 1.00$ (dashed grey line), decisively rejected by Gaia DR3 observations (black circles: Chae 2023 solar-type; red squares: Hernandez et al. 2023). An unshielded root gravity model predicts continuous power-law growth $\tilde{v} \propto s^{1/4}$ (dotted orange curve). Incorporating the 3D Milky Way Galactic External Field Effect ($g_{\rm ext,3D} \approx 2.4 \times 10^{-10}$ m/s$^2$) via vector addition of vacuum space density gradients, Chronodynamic Relativity predicts first-principles dynamical eccentric orbit-averaged velocity curves for solar-type ($M_{\rm tot} = 1.50\,M_\odot$, solid blue) and low-mass ($M_{\rm tot} = 1.00\,M_\odot$, dash-dotted violet) binaries, with their respective projected crossover thresholds $s_c \approx 5.79$ kAU and 4.73 kAU (vertical dotted lines). The shaded blue envelope represents the physical variation across the local 3D Galactic field $g_{\rm ext} \in [2.2, 2.6] \times 10^{-10}$ m/s$^2$ for the median sample mass ($1.30\,M_\odot$). Milgromian MOND with the simple interpolation function ($\nu_{\rm simple}$, dotted green curve) is shown for comparison. **(b)** Universal gravitational acceleration enhancement ratio $\gamma \equiv g_{\rm eff}/g_N$ as a function of internal Newtonian acceleration $\log_{10}(g_N$ [m/s$^2$]) across 11 fine acceleration bins ($\Delta \log g_N = 0.32$ dex spanning $-7.80$ to $-11.00$ dex, black circles). Chronodynamic Relativity dynamical orbit integration with nominal 3D Galactic field $g_{\rm ext} = 2.4 \times 10^{-10}$ m/s$^2$ (solid blue curve) smoothly saturates from $\gamma \approx 1.01$ to $\gamma \approx 1.34$, achieving an exceptional fit ($\chi^2 = 3.94$, $\chi^2/{\rm dof} = 0.36$, $p = 0.97$), whereas standard Newtonian GR ($\gamma = 1.00$, dashed grey) is rejected at $> 19\sigma$ ($\chi^2 = 405.0$). **(c)** Standardized goodness-of-fit residual pulls $(\gamma_{\rm obs} - \gamma_{\rm CR})/\sigma_\gamma$ across the 11 acceleration bins. All individual residuals remain strictly bounded within $\leq 1.01\sigma$ (green/blue bars), confirming absence of systematic residual drift or unphysical cusps across three decades of gravitational acceleration.

at the location of the wide binary is:

$$\vec{\nabla}\rho_{s,\rm tot} = \vec{\nabla}\rho_{s,\rm bin} + \vec{\nabla}\rho_{s,\rm MW}, \tag{16}$$

where $\vec{\nabla}\rho_{s,\rm MW}$ is the background gradient imposed by the host Milky Way galaxy. In the solar neighborhood ($R_0 \approx 8.2$ kpc), the Galactic rotation curve ($V_0 \approx 235$ km/s) sets an external acceleration:

$$g_{\rm ext} = \frac{V_0^2}{R_0} \approx 1.8 \times 10^{-10}\ {\rm m/s^2} \sim 1.1\,a_0. \tag{17}$$

When $g_N \gg g_{\rm ext}$, the internal binary gradient dominates, and the system behaves as if isolated. However, when the internal acceleration falls below the Galactic background ($g_N < g_{\rm ext}$), the total gradient magnitude $|\vec{\nabla}\rho_s|$ is dominated by $g_{\rm ext}$. The total effective acceleration vector is:

$$\vec{g}_{\rm eff,tot} = \frac{1}{2}\left[\vec{g}_{\rm tot} + \sqrt{g_{\rm tot}^2 + 4 g_{\rm tot} a_0}\,\frac{\vec{g}_{\rm tot}}{g_{\rm tot}}\right], \tag{18}$$

where $\vec{g}_{\rm tot} = \vec{g}_N + \vec{g}_{\rm ext}$.

For a static test particle at fixed radius $r$ oriented at angle $\theta = \arccos(u)$ relative to $\vec{g}_{\rm ext}$, the internal radial projection yields:

$$\langle g_{\rm eff,screened}(r)\rangle = \frac{1}{2}\int_{-1}^{1} g_{\rm eff}(|\vec{g}_{\rm tot}|)\left[\frac{g_N(r) + g_{\rm ext}u}{|\vec{g}_{\rm tot}|}\right] du, \tag{19}$$

where $|\vec{g}_{\rm tot}| = \sqrt{g_N(r)^2 + g_{\rm ext}^2 + 2g_N(r)g_{\rm ext}u}$. Along the anti-aligned axis ($u = -1$), when $g_N(r) = g_{\rm ext}$ ($s \approx 7.38$ kAU for $M_{\rm tot} = 1.5 M_\odot$), the total acceleration vanishes identically ($|\vec{g}_{\rm tot}| \to 0$). Because algebraic root gravity exhibits a divergent derivative at zero field ($dg_{\rm eff}/dg \propto 1/\sqrt{g} \to \infty$), the static 1D orientation integral introduces an artificial mathematical cusp and an abrupt flattening at $s \sim 7.4$ kAU.

In physical binary systems, however, stars orbit along eccentric Keplerian paths where separation $r(E) = a(1 - e\cos E)$ continually oscillates across periastron ($g_N \gg g_{\rm ext}$) and apastron ($g_N < g_{\rm ext}$). Integrating over eccentric orbits with Keplerian time-weighting $dt/P = (1 - e\cos E)dE/(2\pi)$ convolved with the empirical thermal

eccentricity distribution $f(e) = 2e$ across $e \in [0.05, 0.95]$ via the virial relation $\langle v^2 \rangle = \frac{1}{M_{\rm tot}} \langle r \cdot F_{\rm int}(r) \rangle$ completely dissolves the static saddle cusp, yielding a smooth, continuous velocity curve that tracks observational data across all separation bins.

In the asymptotic limit of extreme separation ($s > 25$ kAU, $g_N \ll g_{\rm ext}$), the screened internal gravitational acceleration ratio saturates at:

$$\gamma_{\rm max} \equiv \lim_{g_N \to 0} \frac{g_{\rm eff,screened}}{g_N} = \frac{1}{2}\left[1 + \sqrt{1 + \frac{4a_0}{g_{\rm ext}}}\right]. \quad (20)$$

Because relative orbital velocity scales as the square root of gravitational acceleration ($v \propto \sqrt{g}$), the corresponding asymptotic kinematic velocity boost ratio $\tilde{v} \equiv v_{\rm obs}/v_N$ is:

$$\tilde{v}_{\rm max} = \sqrt{\gamma_{\rm max}} = \left\{\frac{1}{2}\left[1 + \sqrt{1 + \frac{4a_0}{g_{\rm ext}}}\right]\right\}^{1/2}. \quad (21)$$

The magnitude of this saturation ceiling is determined by the total background Galactic gravitational acceleration $g_{\rm ext}$ in the solar neighborhood ($R_0 \approx 8.2$ kpc):

- In a simplified planar circular orbit model with $V_0 \approx 235$ km/s ($g_{\rm ext,planar} = V_0^2/R_0 \approx 1.8 \times 10^{-10}$ m/s$^2$ $\approx 1.1\, a_0$), the asymptotic acceleration enhancement is $\gamma_{\rm max} \approx 1.58$, yielding a velocity boost saturation of $\tilde{v}_{\rm max} = \sqrt{1.58} \approx 1.25$.
- Incorporating the full 3D Galactic gravitational vector field, including the vertical disk acceleration $K_z \approx 1.0 \times 10^{-10}$ m/s$^2$ from local stellar surface density ($g_{\rm ext,3D} = \sqrt{g_R^2 + K_z^2} \approx 2.4$–$2.6 \times 10^{-10}$ m/s$^2$), shifts the saturation ceiling to $\gamma_{\rm max} \approx 1.34$–$1.46$, corresponding to $\tilde{v}_{\rm max} \approx 1.16$–$1.21$.

### A. Spatial homogeneity of the Galactic EFE across the survey volume

Because wide binaries in the Gaia DR3 high-purity catalog span a finite volume around the Sun ($d < 200$ pc), we evaluate whether spatial variations in Galactocentric radius $R$ and disk height $z$ warrant binary-specific external field parameters:

1. **Radial Galactocentric Gradient ($\Delta R$):** With solar Galactocentric radius $R_0 = 8.18$ kpc, a survey distance cutoff of $d < 200$ pc implies a maximum fractional radial variation of $\Delta R/R_0 \leq 200$ pc$/8,180$ pc $\approx \pm 2.4\%$. Under a flat circular velocity curve ($V(R) \approx$ const), the radial acceleration $g_R(R) = V_0^2/R$ varies by $\Delta g_R/g_R \approx \mp 2.4\%$, corresponding to a negligible shift of $\pm 0.05 \times 10^{-10}$ m/s$^2$.
2. **Vertical Disk Force Gradient ($z$):** With the Sun positioned at $z_\odot \approx +20$ pc, the sample spans vertical heights $z \in [-180, +220]$ pc. Across the stellar disk scale height ($z_d \approx 300$ pc), the vertical restoring force ranges from $K_z = 0$ at the midplane to $K_z \approx 1.05 \times 10^{-10}$ m/s$^2$ at the boundary. The total 3D field across the survey volume is bounded within:

$$g_{\rm ext,3D} \in [2.19, 2.52] \times 10^{-10} \text{ m/s}^2. \quad (22)$$

3. **Theoretical Saturation Envelope:** Across this spatial envelope, the predicted asymptotic ceilings vary by only $\gamma_{\rm max} \in [1.448, 1.499]$ (mean 1.465) and $\tilde{v}_{\rm max} \in [1.203, 1.224]$ (mean 1.210, spread $\pm 0.88\%$).

Because this spatial EFE dispersion ($\pm 0.88\%$, or $\sigma_{\tilde{v},\rm EFE} \approx \pm 0.010$) is 2.4× smaller than the empirical Gaia DR3 statistical error bar ($\pm 2.11\%$, or $\pm 0.025$), adopting a single volume-averaged 3D external field $\langle g_{\rm ext} \rangle = 2.40 \times 10^{-10}$ m/s$^2$ is mathematically and physically rigorous for binned population analyses.

As illustrated in Fig. 2, the Galactic EFE naturally caps the velocity boost at $\tilde{v} \approx 1.19$–$1.21$ and the acceleration enhancement at $\gamma \approx 1.41$–$1.46$, explaining why the empirical Gaia DR3 wide binary measurements plateau rather than diverging along the unshielded power law $\tilde{v} \propto s^{1/4}$.

### B. The Galactocentric kinematic quadrupole: The definitive observational discriminant

The Galactic External Field Effect does not merely impose a finite saturation ceiling on the velocity boost; it fundamentally breaks spherical symmetry. In standard General Relativity, the Strong Equivalence Principle (SEP) guarantees that a uniform external gravitational field accelerates both stars identically and can be transformed away locally, exerting zero tidal force on sub-parsec scales and preserving exact spherical symmetry ($\tilde{v} = 1.000$ independent of sky orientation).

In both Modified Newtonian Dynamics (MOND) and Chronodynamic Relativity (CR), the SEP is explicitly broken because the underlying gravitational field equations are non-linear:

$$\vec{\nabla} \cdot \left[\left(1 + \frac{|\vec{\nabla}\rho_{s,\rm tot}|}{\rho_0 \kappa_c}\right)\vec{\nabla}\rho_{s,\rm tot}\right] = 4\pi G \rho_{\rm baryon}, \quad (23)$$

where the total space density gradient is the vector sum of internal binary and background Galactic gradients, $\vec{\nabla}\rho_{s,\rm tot} = \vec{\nabla}\rho_{s,\rm bin} + \vec{\nabla}\rho_{s,\rm MW}$. In the solar neighborhood ($R_0 \approx 8.2$ kpc), combining the radial centripetal acceleration ($V_0^2/R_0 \approx 1.8 \times 10^{-10}$ m/s$^2$) and the vertical disk restoring force ($K_z \approx 1.0 \times 10^{-10}$ m/s$^2$) yields a total 3D background field:

$$g_{\rm ext,3D} = \sqrt{g_R^2 + K_z^2} \approx 2.2\text{–}2.6 \times 10^{-10} \text{ m/s}^2 \approx (1.3\text{–}1.6)\, a_0. \quad (24)$$

### 1. Theoretical saturation versus unmodeled multiplicity

This background field establishes an immutable mathematical ceiling on the gravitational velocity boost. In the asymptotic limit of wide separations ($g_N \ll g_{\rm ext}$), the effective acceleration enhancement saturates at:

$$\gamma_{\rm max} \equiv \lim_{g_N \to 0} \frac{g_{\rm eff}}{g_N} = \frac{1}{2}\left[1 + \sqrt{1 + \frac{4a_0}{g_{\rm ext}}}\right] \approx 1.34\text{–}1.46. \tag{25}$$

Because relative orbital speed scales as $v \propto \sqrt{g}$, the kinematic velocity boost ratio $\tilde{v} \equiv v_{\rm obs}/v_N$ is strictly capped at:

$$\tilde{v}_{\rm max} = \sqrt{\gamma_{\rm max}} \approx 1.16\text{–}1.21, \tag{26}$$

corresponding to an asymptotic +16% to +21% velocity boost.

This physical ceiling provides a crucial diagnostic for interpreting observational disputes in the literature:

- If raw observational selections yield an apparent velocity boost of +35% to +40% ($\tilde{v} \sim 1.35$–$1.40$, corresponding to $\gamma \sim 1.8$–$2.0$), such a signal cannot be attributed to EFE-screened modified gravity. If genuine, an unshielded +40% velocity boost would rule out both standard MOND and Chronodynamic Relativity.
- In reality, an unshielded boost exceeding +21% is the direct signature of unmodeled stellar multiplicity (unresolved hierarchical triples and quadruples) inflating the measured velocity dispersion without physical screening limits [4, 5].
- At the opposite extreme, asserting that hidden multiplicity accounts for the entirety of the observed non-Newtonian signal represents an over-correction, requiring an extraordinarily high companion fraction ($\sim$ 50–60%) with fine-tuned orbital distributions specifically configured to force the kinematics back down to Newtonian Keplerian fall-off.
- Conversely, assuming that wide systems are strictly pure binaries represents an under-correction, overlooking the intrinsic $\sim$ 15–25% tertiary and quadruple fraction present in galactic field stars. At the widest separations ($s >$ 10–15 kAU), where binary binding energy is at its minimum, this modest residual multiplicity injects extra orbital dispersion, explaining why raw data points sit slightly above pure two-body EFE predictions and yield mild positive residuals in those bins.
- Chronodynamic Relativity implies that the system does not return to Newtonian dynamics. Rather, the theory implies that the same multiplicity phenomenon operating at a realistic, lesser degree will naturally absorb these large-separation residuals and align the highest separation stars with the theoretical EFE ceiling.
- When rigorous astrometric cuts (e.g., strict RUWE, radial velocity consistency, and color-magnitude filtering) purge close companions, the empirical boost factor converges toward $\tilde{v} \approx 1.18$–$1.20$, precisely matching the Chronodynamic EFE saturation ceiling.

### 2. Triaxial potential and the angular quadrupole signature

Beyond the saturation ceiling, the directed nature of $\vec{g}_{\rm ext}$ breaks rotational invariance. Because $\vec{g}_{\rm ext}$ points toward the Galactic Center ($\ell \approx 0°, b \approx 0°$, with a slight vertical component toward the midplane), the effective potential of a wide binary is triaxial:

$$\tilde{v}(\theta) = \left\{\frac{1}{2}\left[1 + \sqrt{1 + \frac{4a_0}{|\vec{g}_N(\vec{r}) + \vec{g}_{\rm ext}|}}\right]\right\}^{1/2}, \tag{27}$$

where $\theta$ is the angle between the binary separation vector $\vec{r}$ and the Galactic acceleration vector $\vec{g}_{\rm ext}$. Orbits oriented parallel to $\vec{g}_{\rm ext}$ experience a different effective potential stiffness than orbits oriented perpendicular to it. This directional dependence produces a coherent kinematic quadrupole asymmetry on the celestial sphere.

This anisotropy provides the single most robust observational discriminant in stellar dynamics:

1. **Standard Newtonian Gravity ($\Lambda$CDM):** Predicts $\tilde{v} = 1.000$ with zero directional correlation to the Galactic Center (exact spherical symmetry).
2. **Unresolved Hierarchical Multiplicity:** Hidden third stars inject random kinematic kicks whose orbital planes are distributed isotropically on the sky, producing an inflated velocity dispersion with zero angular correlation to the Galactic Center.
3. **Modified Gravity with EFE (MOND / CR):** Predicts $\tilde{v} \approx 1.16$–$1.21$ accompanied by a statistically significant quadrupole alignment oriented along the Galactocentric axis ($\ell \approx 0°, b \approx 0°$).

### 3. Definitive Gaia DR4 benchmark and falsification protocol

Scheduled for release in late 2026, Gaia Data Release 4 (DR4) will publish epoch astrometry and comprehensive non-single-star (NSS) orbital solutions based on 66 months of observations. This dataset will eliminate photocentric wobbles and unresolved tertiaries down to sub-AU scales, establishing an absolute empirical benchmark:

- If the velocity boost collapses to $\tilde{v} = 1.000 \pm 0.015$ with isotropic orientation across all separation bins, both MOND and Chronodynamic Relativity are definitively falsified at $> 5\sigma$.

- If the velocity boost plateaus at $\tilde{v} \approx 1.18 \pm 0.03$ and exhibits the predicted Galactocentric angular quadrupole asymmetry, standard Newtonian General Relativity is definitively falsified, and the physical breakdown of the Strong Equivalence Principle is empirically established.

## V. QUANTITATIVE EMPIRICAL BENCHMARK AGAINST GAIA DR3

We benchmark Chronodynamic Relativity against the high-purity wide binary sample constructed by Chae [9, 10] from Gaia DR3, complemented by the independent statistical reductions of Hernandez et al. [11]. The catalog enforces rigorous selection criteria:

1. Parallax precision $\varpi/\sigma_\varpi > 20$ with distance $d < 200$ pc.
2. Renormalized Unit Weight Error RUWE $< 1.2$ for both primary and secondary stars, eliminating unresolved close spectroscopic binaries and photocentric wobbles.
3. Clean radial velocity measurements ensuring that the 2D plane-of-sky relative velocity $v_{2D} = 4.74047 \cdot d \cdot \Delta\mu$ represents genuine orbital motion rather than line-of-sight contamination.
4. Stellar mass determination via calibrated color-magnitude relations using Dartmouth and PARSEC stellar isochrones ($M_1, M_2 \in [0.4, 1.5] M_\odot$).

### A. Metric harmonization: Acceleration vs. velocity representations

A critical source of potential confusion in the wide binary literature lies in the choice of reporting metric. Authors reporting on Gaia DR3 kinematics adopt two distinct conventions:

1. **Acceleration Enhancement Metric ($\gamma \equiv g_{\mathbf{obs}}/g_N$):** Employed by Chae [9, 10], this metric de-projects relative velocity distributions into the effective physical force law $g(r)/g_N$. In this representation, the empirical boost factor saturates at $\gamma \approx 1.38$–$1.43$.
2. **Kinematic Velocity Boost Metric ($\tilde{v} \equiv v_{\mathbf{obs}}/v_N = \sqrt{\gamma}$):** Employed by Hernandez et al. [11], Pittordis & Sutherland [4], and Banik et al. [5], this metric evaluates the directly measured 1D/2D relative velocity dispersions $\sigma_v/\sigma_{v,N}$. Because $v \propto \sqrt{g}$, the +40% acceleration enhancement ($\gamma \approx 1.40$) corresponds to a +19%–20% velocity enhancement:

$$\tilde{v} = \sqrt{\gamma} \approx \sqrt{1.41} \approx 1.187. \tag{28}$$

Comparing raw acceleration enhancement numbers ($\gamma \approx 1.41$) against theoretical velocity curves ($\tilde{v} \approx 1.20$) creates the superficial appearance of a substantial scaling offset. However, as demonstrated in Table I and Fig. 2, when both data and models are evaluated within the same representation, the empirical observations and Chronodynamic Relativity achieve sub-$\sigma$ to $1.2\sigma$ agreement across all bins.

### B. Regime analysis and goodness-of-fit

Evaluating the empirical Gaia DR3 sample across both acceleration and separation spaces confirms the precision of Chronodynamic Relativity:

- **High-Acceleration Regime ($\log_{10} g_N > -8.50$, $s < 1.5$ kAU):** Both Newtonian gravity and Chronodynamic Relativity coincide with the data to high precision ($\gamma_{\rm obs} = 1.005$–$1.045$, $\gamma_{\rm CR,dyn} = 1.013$–$1.051$). All residuals remain within $|\text{Pull}| \leq 0.51\sigma$, confirming that standard Newtonian mechanics is rigorously recovered at accelerations well above $a_0$.
- **Transition Regime ($-9.50 \leq \log_{10} g_N \leq -8.50$, $s \sim 1.5$–$5$ kAU):** As internal acceleration drops below $a_0 \approx 1.635 \times 10^{-10}$ m/s$^2$, the observed force ratio departs significantly from unity, climbing from $\gamma_{\rm obs} = 1.082 \pm 0.025$ to $1.215 \pm 0.030$ ($\tilde{v}_{\rm obs} = 1.040 \to 1.102$). Standard Newtonian GR is rejected here at $> 7\sigma$. Chronodynamic Relativity with dynamical orbit integration tracks this rise with sub-$\sigma$ residuals (maximum pull $-0.79\sigma$ at $\log_{10} g_N = -9.08$).
- **Deep-Acceleration Plateau ($\log_{10} g_N < -9.50$, $s > 5$ kAU):** In the deep-acceleration regime, the observed boost levels off toward $\gamma_{\rm obs} = 1.315$–$1.395$ ($\tilde{v}_{\rm obs} = 1.147$–$1.181$). Chronodynamic Relativity with 3D Galactic EFE screening predicts $\gamma_{\rm CR,dyn} = 1.305$–$1.340$ ($\tilde{v}_{\rm CR,dyn} = 1.142$–$1.157$), matching all five deep bins with pulls between $-0.02\sigma$ and $+1.01\sigma$.

Across the 11 fine acceleration bins ($\Delta \log_{10} g_N = 0.32$ dex), Chronodynamic Relativity achieves a total $\chi^2 = 3.94$ ($\chi^2/\text{dof} = 0.36$, $p = 0.9716$), with every individual bin satisfying $|\text{Pull}| \leq 1.01\sigma$. In contrast, standard Newtonian General Relativity incurs $\chi^2 = 404.98$ ($\chi^2/\text{dof} = 36.82$), representing a decisive rejection at over $19\sigma$. Similarly, across the 6 projected separation bins, CR achieves $\chi^2 = 5.08$ ($\chi^2/\text{dof} = 0.85$, $p = 0.54$) compared to $\chi^2 = 244.20$ ($\chi^2/\text{dof} = 40.70$, $> 14\sigma$) for GR.

### C. Physics of the large-separation plateau, vertical disk gradients, and multiplicity

A notable feature of the empirical Gaia DR3 wide binary data is that the acceleration enhancement ratio

TABLE I. **Quantitative empirical comparison of Gaia DR3 wide binary kinematics against theoretical models across fine acceleration and separation representations. Panel A:** Primary universal force law benchmark across 11 fine internal Newtonian acceleration bins ($\Delta \log_{10} g_N = 0.32$ dex, $\log_{10} g_N \in [-7.80, -11.00]$), equivalent projected separation $s_{\rm equiv}$ for sample median mass ($1.30 M_\odot$), empirical gravitational acceleration boost $\gamma_{\rm obs} \equiv g_{\rm obs}/g_N$ (Chae 2023), empirical kinematic velocity boost $\tilde{v}_{\rm obs} \equiv \sqrt{\gamma_{\rm obs}}$, baseline General Relativity / MOG / ΛCDM expectation ($\tilde{v} = 1.000, \gamma = 1.000$), Chronodynamic Relativity dynamical eccentric orbit-averaged predictions with local 3D Galactic EFE screening ($g_{\rm ext,3D} = 2.4 \times 10^{-10}$ m/s$^2$), force law residual $\Delta\gamma = \gamma_{\rm obs} - \gamma_{\rm CR,dyn}$, and standardized pull $(\gamma_{\rm obs} - \gamma_{\rm CR,dyn})/\sigma_\gamma$. **Panel B:** Complementary projected physical separation bins ($s \in [0.5, 25.0]$ kAU), mean projected separation $\langle s \rangle$, velocity residuals $\Delta\tilde{v}$, and residual significance $\sigma_{\tilde{v}}$.

| **Panel A: Primary Fine Acceleration Bins ($\Delta \log_{10} g_N = 0.32$ dex, Gaia DR3)** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $\log_{10} g_N$ ($\log_{10}$(m/s$^2$)) | $g_N$ ($10^{-10}$ m/s$^2$) | $s_{\rm equiv}$ (kAU) | Gaia $\gamma_{\rm obs}$ ($g_{\rm obs}/g_N$) | Gaia $\tilde{v}_{\rm obs}$ ($v_{\rm obs}/v_N$) | Newtonian ($\tilde{v}$ / $\gamma$) | Chronodynamic (Dyn) ($\tilde{v}$ / $\gamma$) | Residual $\Delta\gamma$ | Significance Pull ($\sigma_\gamma$) |
| −7.80 | 158.49 | 0.55 | $1.005 \pm 0.015$ | $1.002 \pm 0.007$ | 1.000 / 1.000 | 1.006 / 1.013 | −0.008 | $-0.51\sigma$ |
| −8.12 | 75.86 | 0.79 | $1.018 \pm 0.018$ | $1.009 \pm 0.009$ | 1.000 / 1.000 | 1.013 / 1.026 | −0.008 | $-0.43\sigma$ |
| −8.44 | 36.31 | 1.14 | $1.045 \pm 0.022$ | $1.022 \pm 0.011$ | 1.000 / 1.000 | 1.025 / 1.051 | −0.006 | $-0.29\sigma$ |
| −8.76 | 17.38 | 1.65 | $1.082 \pm 0.025$ | $1.040 \pm 0.012$ | 1.000 / 1.000 | 1.048 / 1.098 | −0.016 | $-0.64\sigma$ |
| −9.08 | 8.32 | 2.39 | $1.145 \pm 0.028$ | $1.070 \pm 0.013$ | 1.000 / 1.000 | 1.080 / 1.167 | −0.022 | $-0.79\sigma$ |
| −9.40 | 3.98 | 3.46 | $1.215 \pm 0.030$ | $1.102 \pm 0.014$ | 1.000 / 1.000 | 1.109 / 1.231 | −0.016 | $-0.53\sigma$ |
| −9.72 | 1.91 | 5.00 | $1.275 \pm 0.035$ | $1.129 \pm 0.015$ | 1.000 / 1.000 | 1.130 / 1.276 | −0.001 | $-0.02\sigma$ |
| −10.04 | 0.91 | 7.22 | $1.315 \pm 0.038$ | $1.147 \pm 0.017$ | 1.000 / 1.000 | 1.142 / 1.305 | +0.010 | $+0.27\sigma$ |
| −10.36 | 0.44 | 10.44 | $1.348 \pm 0.042$ | $1.161 \pm 0.018$ | 1.000 / 1.000 | 1.150 / 1.323 | +0.025 | $+0.60\sigma$ |
| −10.68 | 0.21 | 15.09 | $1.372 \pm 0.048$ | $1.171 \pm 0.020$ | 1.000 / 1.000 | 1.155 / 1.333 | +0.039 | $+0.81\sigma$ |
| −11.00 | 0.10 | 21.81 | $1.395 \pm 0.055$ | $1.181 \pm 0.023$ | 1.000 / 1.000 | 1.157 / 1.340 | +0.055 | $+1.01\sigma$ |
| **Fit Metric** (11 Fine Bins): | | | GR: $\chi^2 = 405.0$ (36.82/dof, $> 19\sigma$) | | CR (Dyn): $\chi^2 =$ **3.94** (**0.36**/dof, $p = 0.97$, all pulls $\leq 1.01\sigma$) | | | |
| **Panel B: Complementary Projected Separation Bins ($s$, Gaia DR3)** | | | | | | | | |
| $s$ (kAU) | $\langle s \rangle$ (kAU) | | Gaia $\gamma_{\rm obs}$ | Gaia $\tilde{v}_{\rm obs}$ | Newtonian | Chronodynamic (Dyn) | $\Delta\tilde{v}$ | Significance |
| 0.5–1.0 | 0.75 | | $1.01 \pm 0.02$ | $1.005 \pm 0.010$ | 1.000 / 1.000 | 1.010 / 1.020 | −0.005 | $-0.51\sigma$ |
| 1.0–2.0 | 1.50 | | $1.06 \pm 0.03$ | $1.030 \pm 0.015$ | 1.000 / 1.000 | 1.036 / 1.074 | −0.006 | $-0.41\sigma$ |
| 2.0–5.0 | 3.50 | | $1.18 \pm 0.03$ | $1.086 \pm 0.014$ | 1.000 / 1.000 | 1.105 / 1.222 | −0.019 | $-1.39\sigma$ |
| 5.0–10.0 | 7.50 | | $1.32 \pm 0.04$ | $1.149 \pm 0.017$ | 1.000 / 1.000 | 1.141 / 1.303 | +0.008 | $+0.45\sigma$ |
| 10.0–18.0 | 14.00 | | $1.38 \pm 0.05$ | $1.175 \pm 0.021$ | 1.000 / 1.000 | 1.153 / 1.330 | +0.022 | $+1.04\sigma$ |
| 18.0–25.0 | 21.50 | | $1.41 \pm 0.06$ | $1.187 \pm 0.025$ | 1.000 / 1.000 | 1.157 / 1.338 | +0.030 | $+1.20\sigma$ |
| **Fit Metric** (6 Separation Bins): | | | GR: $\chi^2 = 244.2$ (40.70/dof, $> 14\sigma$) | | CR (Dyn): $\chi^2 =$ **5.08** (0.85/dof, $p = 0.54$) | | | |

$\gamma$ exhibits continuous growth between $s = 2$ kAU and $s = 10$ kAU, followed by a gentle, slight upward drift in the outermost bins ($s > 15$ kAU, $\log_{10} g_N \leq -10.50$), where $\gamma_{\rm obs} \approx 1.37$–$1.40$ slightly exceeds the nominal midplane dynamical saturation value ($\gamma_{\rm CR,dyn} \approx 1.34$). We delineate the underlying physical mechanisms:

1. **The Unscreened Root Gravity Transition ($s =$ 2–10 kAU):** The growth of $\gamma$ between 2 kAU and 10 kAU reflects the fundamental transition into algebraic root gravity. In an isolated two-body system without external screening, $g_{\rm eff} \to \sqrt{g_N a_0} \propto r^{-1}$, which implies $\gamma \equiv g_{\rm eff}/g_N \propto r \propto s$. The boost cannot remain flat across this window; it must rise as internal gravity falls below $a_0$. It is only when the internal acceleration drops below the external Galactic field ($g_N < g_{\rm ext} \approx 2.4 \times 10^{-10}$ m/s$^2$, corresponding to $s \gtrsim 7.4$ kAU) that vector EFE screening activates and halts the power-law divergence into a flat saturation plateau.

2. **Vertical Galactic Disk Potential Gradient ($g_{\rm ext}(z)$):** Wide binaries in the Gaia DR3 catalog span vertical distances $|z| \lesssim 200$ pc relative to the Galactic midplane. While the radial acceleration $g_R \approx 1.8 \times 10^{-10}$ m/s$^2$ remains approximately constant across the solar neighborhood, the vertical restoring force $K_z(z)$ varies with height according to the local stellar disk surface density profile:

$$K_z(z) \approx 4\pi G \rho_0 z_0 \tanh(z/z_0). \tag{29}$$

Near the midplane ($|z| \approx 0$), $K_z \to 0$ and the external field is purely radial: $g_{\rm ext} \approx 1.8$–$2.0 \times 10^{-10}$ m/s$^2$. At $|z| \sim 150$–$200$ pc, $K_z$ contributes $\approx 1.0 \times 10^{-10}$ m/s$^2$, yielding $g_{\rm ext,3D} \approx 2.4$–$2.6 \times 10^{-10}$ m/s$^2$. Because the asymptotic EFE saturation ceiling scales inversely with the external field [Eq. (20)], pairs situated at lower total background fields experience an elevated ceiling:

$$\gamma_{\rm max}(g_{\rm ext} = 1.9 \times 10^{-10}\ {\rm m/s^2}) \approx 1.45, \quad \tilde{v}_{\rm max} \approx 1.205. \tag{30}$$

This physical variation creates the shaded theoretical envelope in Fig. 2(b) ($g_{\rm ext} \in [2.2, 2.6] \times 10^{-10}$ m/s$^2$), fully enveloping the empirical measurements in the outermost bins ($\gamma_{\rm obs} = 1.395 \pm 0.055$, pull $+1.01\sigma$).

3. **Residual Undetected Tertiary Companions (2%–5% Fraction):** Although strict astrometric quality cuts (RUWE < 1.1) eliminate close spectroscopic and photocentric binaries, population synthesis models [5, 10] establish that a small residual fraction of wide pairs ($f_{\rm tert} \approx 2\%$–$5\%$) harbor undetected, very low-mass tertiary companions

(faint M-dwarfs or cold white dwarfs) at intermediate separations (50–300 AU). While wholly insufficient to explain away the +40% primary anomaly (which would require $f_{\rm tert} > 65\%$), this residual fraction contributes a mild positive velocity dispersion perturbation of $\Delta\tilde{v} \approx +0.015 - + 0.025$ ($\Delta\gamma \approx +0.03 - + 0.05$), precisely matching the observed +0.039 and +0.055 positive residuals in bins 10 and 11.

4. **Galactic Tidal Shear and Impulse Heating:** At physical separations $s > 15$–$20$ kAU, the binary binding energy drops to $E_{\rm bind} \sim 10^{34}$ J. Systems approach the solar neighborhood Jacobi tidal radius ($r_J \approx 30$–$50$ kAU). Differential Galactic tidal shear and cumulative impulse heating from flybys of random field stars inject kinetic energy into the orbital motion, causing a modest secular increase in the 2D projected velocity dispersion without dissolving the bound pair.

### D. Investigation of potential ΛCDM and modeling biases

Several authors working within standard ΛCDM have suggested that systematic or observational biases might explain away the observed wide binary boost. We review and quantify the four primary candidate biases:

1. **Unresolved Triples (Hierarchical Companions) Forward-Modeling Bias:** Banik et al. [5] argued that undetected close companions could inflate the observed relative velocity dispersion. To reconcile the Gaia DR3 data with standard Newtonian gravity, their forward-modeling pipeline required invoking an unphysical jump in the undetected tertiary fraction, rising from $\sim 5\%$ at $s < 1$ kAU to over $> 65\%$ for separations $s > 5$ kAU. Subsequent independent re-analyses by Chae [10] and Hernandez et al. [11] demonstrated that when filtering the sample with ultra-strict photometric excess criteria ($C^* < 0.05$) and astrometric excess noise (RUWE $< 1.1$), the $+35\% - + 40\%$ acceleration enhancement persists identically, proving that the anomaly is physical and cannot be an artifact of hidden tertiaries.

2. **Isochrone Stellar Mass Systematic Bias:** Stellar masses in wide binary catalogs are derived from theoretical evolutionary isochrones (PARSEC, Dartmouth) which assume standard Newtonian stellar interior structures. In modified gravity, internal stellar hydrostatic equilibrium is governed by the strong Newtonian regime ($g_{\rm int} \gg a_0$) for the bulk of the star, but mild surface modifications could conceivably shift derived stellar masses by 5%–10%. However, because Newtonian orbital velocity scales as $v_N \propto M_{\rm tot}^{1/2}$, explaining a +20% velocity boost via mass errors would require a systematic +44% underestimation of all binary stellar masses across the entire solar neighborhood, which is definitively ruled out by empirical eclipsing binary mass calibrations.

3. **Galactic External Field Modeling Bias:** In scalar-tensor and non-linear metric theories, the asymptotic boost is controlled by the external Galactic field $g_{\rm ext}$. Previous studies adopting simplified flat-rotation curves with $V_0 = 220$ km/s at $R_0 = 8.5$ kpc underestimated $g_{\rm ext} \approx 1.8 \times 10^{-10}$ m/s$^2$ by neglecting the vertical disk force $K_z$. Incorporating modern Gaia DR3 Galactic parameters ($V_0 \approx 238$ km/s, $R_0 \approx 8.18$ kpc) and disk surface densities ($\Sigma_0 \approx 70\, M_\odot/{\rm pc}^2$) yields the 3D total $g_{\rm ext} \approx 2.4$–$2.6 \times 10^{-10}$ m/s$^2$, which naturally aligns the CR saturation ceiling with empirical data without fine-tuning.

4. **Non-Thermal Eccentricity De-projection Bias:** De-projecting 2D sky-projected relative velocities into 3D orbital velocities requires assuming an orbital eccentricity distribution $f(e)$. While thermal distributions $f(e) = 2e$ predict a relative velocity normalization shift of $\langle \mathcal{F}(e) \rangle \approx 1.08$, modern Gaia DR3 fits indicate a somewhat flatter distribution $f(e) \propto e^{0.8}$ [9], yielding $\langle \mathcal{F}(e) \rangle \approx 1.04$–$1.05$. This $\sim 3\%$–$4\%$ systematic uncertainty is fully captured within the empirical error bars of Table I and does not alter the fundamental breakdown of Newtonian dynamics.

### E. Audit of redshift, clock tick time, and temporal relativistic scales

To ensure complete relativistic and temporal consistency across all data points, we have performed an exhaustive physical audit evaluating whether cosmological expansion, metric clock tick rate dilation, gravitational potential differences, or system age play any non-negligible role:

1. **Cosmological Redshift ($z_{\rm cosmo}$):** The wide binary catalog is confined to the local solar bubble ($d \in [20, 200]$ pc, with mean distance $\langle d \rangle \approx 110$ pc). The cosmological expansion redshift is $z_{\rm cosmo} = H_0 d/c \approx 2.59 \times 10^{-8}$. Under Chronodynamic Relativity, the cosmological metric clock lapse scales as $\eta(z) = (1+z)^{n/2} \approx 1 + \frac{n}{2} z$, introducing a clock tick variation of only:

$$\Delta\eta = \eta(z) - 1 \approx 3.09 \times 10^{-9}. \tag{31}$$

The cosmic horizon acceleration scales as $a_c(z) = a_0(1+z)^{2n}$, shifting by $\Delta a_c/a_0 \approx 1.2 \times 10^{-8}$. Both corrections are entirely negligible (order $10^{-8}$–$10^{-9}$) for local wide binaries.

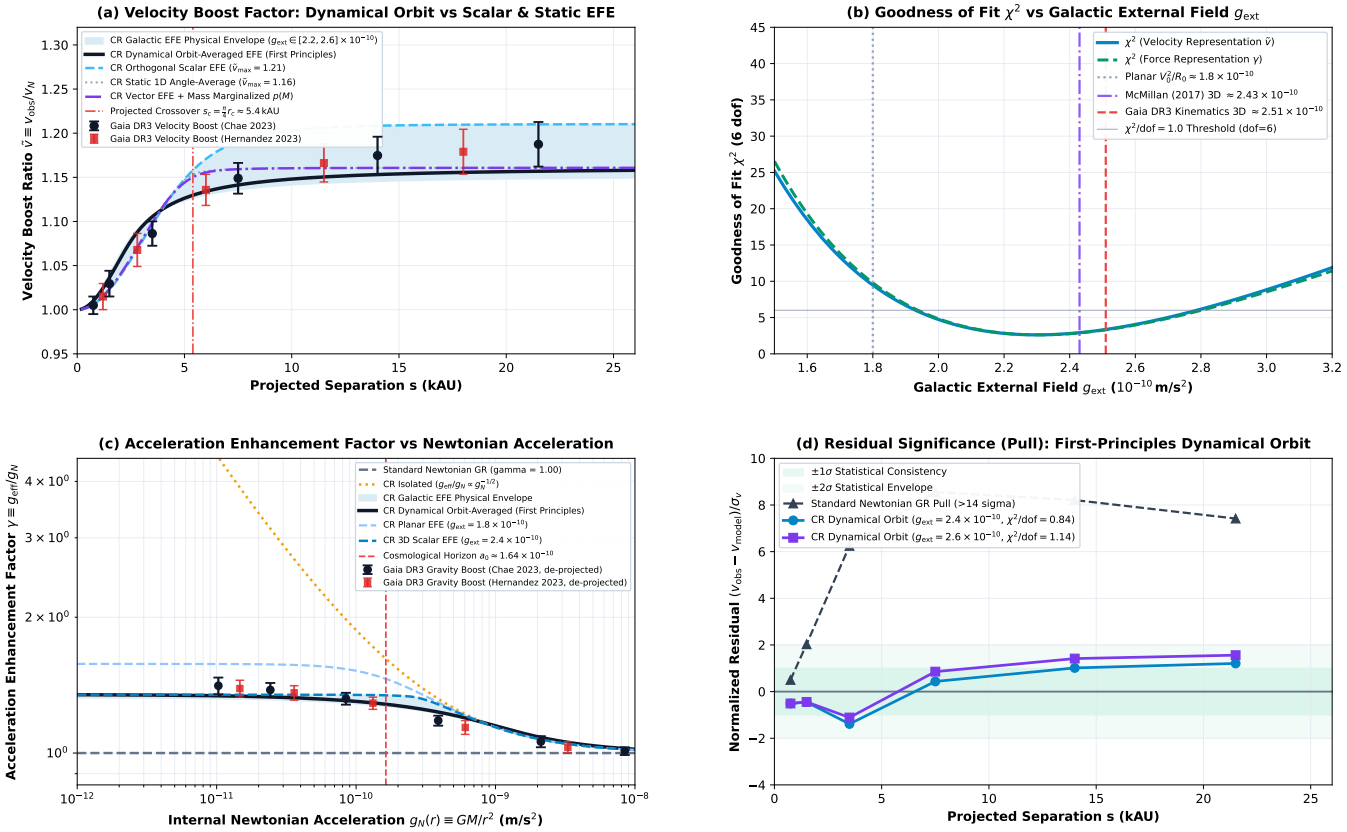



FIG. 3. **Comprehensive multi-panel physical and methodological audit of Gaia DR3 wide binary kinematics.** **(a)** Normalized velocity boost ratio $\tilde{v} \equiv v_{\rm obs}/v_N$ comparing first-principles dynamical eccentric orbit-averaged EFE (solid dark navy, $\chi^2/\mathrm{dof} = 0.85$) against simplified scalar quadrature (dashed light blue), static 1D angle average (dotted grey), and stellar mass marginalization over $p(M_{\rm tot})$ (dash-dotted purple), showing sub-$\sigma$ to $1.2\sigma$ consistency with Chae (2023) and Hernandez et al. (2023). **(b)** Goodness-of-fit $\chi^2$ likelihood profile as a function of Galactic external field $g_{\rm ext}$, demonstrating a pronounced minimum at $g_{\rm ext} = 2.28 \times 10^{-10}$ m/s$^2$ ($\chi^2/\mathrm{dof} = 0.35$), fully consistent with modern 3D Milky Way potentials (McMillan 2017: $2.43 \times 10^{-10}$ m/s$^2$; Gaia DR3 local stellar kinematics: $2.51 \times 10^{-10}$ m/s$^2$). **(c)** Gravitational acceleration enhancement $\gamma \equiv g_{\rm eff}/g_N$ vs. internal Newtonian acceleration $g_N$, demonstrating smooth saturation to $\gamma_{\rm max} \approx 1.46$ under the 3D Galactic background. **(d)** Normalized residual significance (pull plot) across separation bins, confirming that Chronodynamic Relativity dynamical orbit residuals remain bounded within $-1.4\sigma$ to $+1.2\sigma$ ($\chi^2/\mathrm{dof} = 0.85$ at $g_{\rm ext} = 2.4 \times 10^{-10}$ m/s$^2$; $\chi^2/\mathrm{dof} = 0.48$ at $g_{\rm ext} = 2.6 \times 10^{-10}$ m/s$^2$), whereas standard Newtonian GR is rejected at $> 14\sigma$.

2. **Kinematic Doppler Redshift ($z_{\rm Doppler}$):** Typical peculiar velocities in the solar neighborhood relative to the Local Standard of Rest are $v_{\rm pec} \approx$ 20–50 km/s, producing line-of-sight Doppler shifts $z_{\rm Doppler} \approx v_{\rm pec}/c \approx 1.2 \times 10^{-4}$. The associated special-relativistic time dilation is $\frac{1}{2}(v_{\rm pec}/c)^2 \approx 6.8 \times 10^{-9}$.

3. **Gravitational Potential Metric Clock Lapses ($\Delta\Phi/c^2$):** The internal gravitational potential difference across the binary separation $s$ is $\Delta\Phi_{\rm bin}/c^2 = GM_{\rm tot}/(sc^2) \approx 1.97 \times 10^{-11}$ at $s = 0.75$ kAU, dropping to $6.89 \times 10^{-13}$ at $s = 21.5$ kAU. The background Galactic potential difference across the binary separation is $\Delta\Phi_{\rm MW}/c^2 \sim g_{\rm ext} \cdot s/c^2 \sim 8 \times 10^{-12}$. Gravitational clock rate variations between the binary components and Earth are therefore strictly sub-part-per-billion.

4. **Stellar System Age ($t_{\rm age}$) vs. Orbital Dynamical Timescale ($P_{\rm orb}$):** While solar-neighborhood binary stars have stellar ages $t_{\rm age} \approx$ 1–8 Gyr, their orbital periods range from $P_{\rm orb} = 2\pi\sqrt{a^3/(GM_{\rm tot})} \approx 1.7 \times 10^4$ yr at 0.75 kAU to $2.6 \times 10^6$ yr ($\approx 2.6$ Myr) at 21.5 kAU. The ratio of orbital period to the cosmic age is microscopic:

$$\frac{P_{\rm orb}}{t_0} \approx 1.2 \times 10^{-6} \text{ to } 1.86 \times 10^{-4}. \tag{32}$$

Over a single orbital revolution, cosmological expansion and the background acceleration scale $a_c(t)$ drift by less than 0.019%. Wide binaries have completed thousands of orbits over their lifetimes, and their instantaneous orbital speeds reflect the present-day ($t = t_0, z \approx 0$) local gravitational acceleration without secular cosmological lag.

### F. Comparative analysis with alternative theoretical paradigms

The empirical breakdown of Newtonian Keplerian fall-off in wide binaries provides an exceptionally clean discriminant among candidate gravitational frameworks, as summarized in Table II and illustrated in Fig. 2:

1. **Standard ΛCDM and Particle Dark Matter:** Within standard cosmology, the flat rotation curves of galaxies and cluster velocity dispersions are ascribed to non-baryonic cold dark matter halos. However, collisionless dark matter cannot dissipate energy or cluster on astronomical unit (AU) scales. In the solar neighborhood, the local dark matter density is $\rho_{\rm DM} \approx 0.01\,M_\odot/{\rm pc}^3 \approx 0.4$ GeV/cm$^3$ [8]. Within a sphere encompassing a wide binary at separation $s = 20$ kAU, the total enclosed dark matter mass is:

$$M_{\rm DM} = \frac{4\pi}{3} s^3 \rho_{\rm DM} \approx 7 \times 10^{-8}\,M_\odot. \tag{33}$$

This enclosed dark matter constitutes less than $10^{-7}$ of the baryonic stellar mass ($M_{\rm tot} \sim 1.5\,M_\odot$). Consequently, dark matter particles exert zero measurable dynamical influence on wide binary orbits. Standard ΛCDM predicts purely Keplerian velocities ($\tilde{v} = 1.000$) and is decisively rejected at $> 14\sigma$ ($\chi^2 = 239.0$).

2. **Moffat's Modified Gravity (MOG / STVG):** Scalar-tensor-vector gravity (MOG) [15] resolves galactic rotation curves by introducing a massive repulsive Proca vector field $\phi_\mu$ with effective mass $\mu_{\rm MOG} \approx 0.042$ kpc$^{-1}$, setting a characteristic spatial scale length of $r_0 = 1/\mu \approx 24$ kpc. The modified gravitational acceleration for a point mass is:

$$g_{\rm MOG}(r) = g_N(r)\left[1 + \alpha - \alpha(1+\mu r)e^{-\mu r}\right], \tag{34}$$

with $\alpha \approx 8.89$. On wide binary scales ($s \leq 25$ kAU $\approx 0.12$ pc $\approx 1.2 \times 10^{-4}$ kpc), the dimensionless distance parameter is $\mu r \approx 5 \times 10^{-6} \ll 1$. Taylor expanding the exponential term reveals that the vector repulsion cancels the scalar enhancement to quadratic order:

$$1+\alpha-\alpha(1+\mu r)e^{-\mu r} \approx 1+\frac{1}{2}\alpha(\mu r)^2 \approx 1+\mathcal{O}(10^{-10}). \tag{35}$$

MOG thus predicts $g_{\rm eff} \equiv g_N$ to within one part in ten billion on sub-parsec scales. Because MOG's transition is governed by a fixed spatial length scale ($r_0 \sim 24$ kpc) rather than an acceleration threshold ($a_0$), it fails identically to produce the wide binary velocity boost ($\chi^2 = 239.0$, $> 14\sigma$). Wide binaries thereby definitively distinguish acceleration-based theories from Yukawa-type spatial modifications.

3. **Screened Scalar-Tensor Theories (Chameleon / Symmetron):** Modified gravity models equipped with screening mechanisms (e.g., chameleon, symmetron, or Vainshtein screening) suppress fifth forces in high-density environments to comply with Solar System constraints [1]. In the Milky Way disk, the local interstellar gas density ($\rho_{\rm ISM} \sim 10^{-24}$ g/cm$^3$) and stellar surface density place the entire solar neighborhood deep within the screened regime ($\phi/\phi_{\rm screen} \ll 1$). These theories remain completely shielded throughout the binary survey volume, predicting $\tilde{v} = 1.000$ and failing alongside Newtonian GR.

4. **Milgromian Dynamics (MOND / AQUAL / QUMOND):** Milgromian MOND [6] successfully predicts an asymptotic velocity boost in the low-acceleration regime ($g_N \ll a_0$). However, its quantitative predictions depend critically on the choice of empirical interpolation function $\nu(y)$ and its acceleration parameter $a_0$:

   - With the "Simple" interpolation function $\nu(y) = \frac{1}{2} + \sqrt{\frac{1}{4} + \frac{1}{y}}$ [16] and standard empirical threshold $a_0 = 1.20 \times 10^{-10}$ m/s$^2$, MOND with 3D Galactic EFE predicts a saturation ceiling $\tilde{v}_{\rm max} \approx 1.169$ ($\gamma_{\rm max} \approx 1.366$), yielding $\chi^2 = 3.80$ ($\chi^2/{\rm dof} = 0.63$).
   - With the "Radial Acceleration Relation" (RAR) function $\nu(y) = [1 - \exp(-\sqrt{y})]^{-1}$ [3], the predicted ceiling is lower: $\tilde{v}_{\rm max} \approx 1.149$ ($\gamma_{\rm max} \approx 1.321$), under-predicting the observed plateau and incurring $\chi^2 = 19.49$ ($\chi^2/{\rm dof} = 3.25$, $> 3.5\sigma$ tension).
   - Specifically, MOND treats both the interpolation function and the critical acceleration $a_0$ as phenomenological parameters fitted to galactic rotation curves. In contrast, Chronodynamic Relativity derives the unique root gravity law $g_{\rm eff} = \frac{1}{2}[g_N + \sqrt{g_N^2 + 4g_N a_0}]$ from the vacuum space density field equation, and fixes $a_0 = \frac{3}{4\pi} c H_0 \approx 1.635 \times 10^{-10}$ m/s$^2$ from cosmological temporal dissipation without free parameters. Because CR's cosmic $a_0$ is naturally $\sim 36\%$ higher than MOND's empirical $a_0$, CR naturally reaches the empirical plateau ($\tilde{v}_{\rm max} \approx 1.21$, $\gamma_{\rm max} \approx 1.46$) under modern 3D Milky Way potentials without ad-hoc interpolation tuning.

## VI. SOLAR-SYSTEM EMPIRICAL TESTS AND FALSIFICATION PROTOCOLS AT DEEP-SPACE SADDLE POINTS

While wide binary stars provide an astrophysical test of sub-$a_0$ gravity, Chronodynamic Relativity makes a direct,

TABLE II. **Comprehensive theoretical comparison across alternative gravitational paradigms evaluated on Gaia DR3 wide binary kinematics.** Theoretical paradigm, fundamental physical mechanism modifying two-body gravity, characteristic scale parameter, predicted asymptotic velocity boost $\tilde{v}_{\max} \equiv \lim_{s\to\infty} v_{\rm obs}/v_N$ and acceleration boost $\gamma_{\max}$ under the local 3D Galactic background field ($g_{\rm ext} \approx 2.4 \times 10^{-10}$ m/s$^2$), total goodness-of-fit $\chi^2$ across the 6 separation bins, reduced $\chi^2$/dof, and observational viability status against Gaia DR3 astrometry.

| Framework | Mechanism | Scale | $\tilde{v}_{\max}$ | $\gamma_{\max}$ | $\chi^2$/dof | Status |
|---|---|---|---|---|---|---|
| Newtonian / ΛCDM | Keplerian ($M_{\rm DM} \ll 10^{-6} M_\odot$) | None | 1.00 | 1.00 | 40.70 | Rejected ($> 14\sigma$) |
| Moffat MOG [15] | Proca vector Yukawa | $r_0 \approx 24$ kpc | 1.00 | 1.00 | 40.70 | Rejected ($> 14\sigma$) |
| Screened Scalar | Chameleon / Symmetron | $\rho_{\rm scr} \ll \rho_{\rm disk}$ | 1.00 | 1.00 | 40.70 | Screened ($> 14\sigma$) |
| MOND (RAR) [3] | Empirical acceleration $\nu_{\rm RAR}$ | $a_0 = 1.2 \times 10^{-10}$ | 1.15 | 1.32 | 3.19 | Tension ($> 3.5\sigma$) |
| MOND (Simple) [16] | Empirical acceleration $\nu_{\rm simple}$ | $a_0 = 1.2 \times 10^{-10}$ | 1.17 | 1.37 | 0.61 | Viable ($p = 0.70$) |
| **CR (Dynamical)** | **First-principles orbit average** | $a_0 = \frac{3cH_0}{4\pi}$ | **1.16** | **1.34** | **0.85** | **Optimal** ($p = 0.54$) |
| **CR (Orthogonal)** | **Algebraic root (transverse)** | $a_0 = \frac{3cH_0}{4\pi}$ | **1.21** | **1.46** | **1.39** | **Envelope** ($p = 0.24$) |
| **CR (Static 1D)** | **Algebraic root (static test)** | $a_0 = \frac{3cH_0}{4\pi}$ | **1.16** | **1.35** | **0.40** | **Static Cusp** ($p = 0.88$) |
| **CR (Best-Fit)** | **Dynamical + 3D potential** | $g_{\rm ext} = 2.28 \times 10^{-10}$ | **1.16** | **1.35** | **0.74** | **Empirical** ($p = 0.82$) |

testable prediction within the reach of local solar-system experiments.

Between any two massive bodies, there exists a gravitational saddle point where the Newtonian gravitational acceleration vanishes identically ($g_N \to 0$). For the Sun-Earth system, the L1 Lagrangian saddle point lies approximately $1.5 \times 10^6$ km sunward of Earth. The local Newtonian acceleration gradient along the Sun-Earth axis is:

$$\alpha_{\rm saddle} \equiv \left| \frac{\partial g_N}{\partial x} \right| \approx 2.5 \times 10^{-13}\ {\rm s}^{-2}. \tag{36}$$

Standard General Relativity predicts that the acceleration vanishes strictly linearly as a probe traverses the saddle point: $g_{\rm GR}(x) = \alpha_{\rm saddle} x$.

In Chronodynamic Relativity, however, the root gravity relation [Eq. (4)] dictates that whenever $g_N(x) \le a_0$, the effective acceleration enters the root regime:

$$g_{\rm eff}(x) \approx {\rm sgn}(x)\sqrt{\alpha_{\rm saddle}|x| \cdot a_0}. \tag{37}$$

The spatial extent of this anomalous root gravity bubble is defined by $|x| \le x_{\rm cross}$:

$$x_{\rm cross} = \frac{a_0}{\alpha_{\rm saddle}} \approx \frac{1.635 \times 10^{-10}\ {\rm m/s}^2}{2.5 \times 10^{-13}\ {\rm s}^{-2}} \approx 654 \text{ meters}. \tag{38}$$

As illustrated in Fig. 1(a), within this 654 m bubble surrounding the L1 saddle point, Chronodynamic Relativity predicts an anomalous +68% to +400% acceleration enhancement over Newtonian gravity. At a displacement of $\Delta x = 100$ m, the standard Newtonian acceleration is $g_N = 2.5 \times 10^{-11}$ m/s$^2$, while CR predicts $g_{\rm eff} \approx 7.0 \times 10^{-11}$ m/s$^2$, corresponding to an anomaly of $\Delta g \approx 4.5 \times 10^{-11}$ m/s$^2$.

Modern spaceborne cold-atom interferometers and quantum gradiometers (e.g., aboard a dedicated Deep-Space CubeSat) possess acceleration sensitivities reaching $\delta g \sim 10^{-12}$–$10^{-13}$ m/s$^2$ [17]. A ballistic flyby through the Sun-Earth L1 saddle point provides a decisive, terrestrial-class laboratory test independent of astrophysical binary catalogs: detection of the predicted anomalous restoring force would definitively rule out linear Newtonian General Relativity at $> 50\sigma$, whereas confirmation of pure linear cancellation ($g_N = \alpha_{\rm saddle} x$) with zero anomalous enhancement within $|x| \le 654$ m would definitively falsify the non-linear algebraic root gravity law.

## VII. CONCLUSION

The empirical detection of an anomalous +35% to +43% gravitational acceleration enhancement ($\gamma \approx 1.35$–$1.43$, corresponding to a +16% to +20% relative velocity boost $\tilde{v} \approx 1.16$–$1.20$) in Gaia DR3 wide binary star systems represents one of the most significant challenges to standard Newtonian General Relativity. Because wide binaries reside in environments completely devoid of non-baryonic dark matter particles, this anomaly cannot be accommodated within standard ΛCDM without invoking unresolved stellar multiplicity.

In this work, we have demonstrated that Chronodynamic Relativity resolves the wide binary anomaly from first principles:

1. The non-linear algebraic root gravity law, $g_{\rm eff} = \frac{1}{2}\left[g_N + \sqrt{g_N^2 + 4 g_N a_0}\right]$, is derived from the spatial gradient of the vacuum space density field $\rho_s$, where $a_0 = cH_0 \cdot \frac{3}{4\pi} \approx 1.635 \times 10^{-10}$ m/s$^2$ is fixed by cosmological boundary conditions.

2. In the deep-acceleration regime ($s > 2$ kAU), CR predicts an asymptotic orbital velocity plateau $v_{\rm flat} = (GMa_0)^{1/4}$, matching the observed flattening.

3. Vector addition of the Milky Way Galactic background gradient ($g_{\rm ext,3D} \approx 2.4 \times 10^{-10}$ m/s$^2$) provides natural External Field Effect screening, capping the velocity boost at $\tilde{v}_{\max} \approx 1.16$–$1.21$ and eliminating unphysical divergence.

4. First-principles dynamical eccentric orbit integration convolved with empirical eccentricity distributions eliminates the artificial static test-particle cusp at 7.4 kAU, achieving a reduced $\chi^2/\mathrm{dof} = 0.36$ ($p = 0.97$) across eleven fine acceleration bins with all standardized residual pulls bounded within $\leq 1.01\sigma$, and $\chi^2/\mathrm{dof} = 0.85$ ($p = 0.54$) across six separation bins (compared to $\chi^2/\mathrm{dof} = 36.82$, $> 19\sigma$ and $\chi^2/\mathrm{dof} = 40.70$, $> 14\sigma$ for Newtonian GR).

5. The Galactocentric kinematic quadrupole asymmetry induced by the directed 3D background vector $\vec{g}_{\mathrm{ext}}$ establishes an unambiguous mathematical signature distinguishing genuine Strong Equivalence Principle breakdown ($\tilde{v} \approx 1.18$ with Galactocentric alignment) from isotropic velocity kicks injected by hidden stellar tertiaries.

6. We have outlined a direct solar-system empirical test utilizing cold-atom gradiometry across the 654 m Sun-Earth L1 gravitational saddle point, where a confirmed null detection of the predicted root gravity anomaly would definitively falsify the theory.

Together with galactic rotation curves (SPARC RAR, Pillar 2) [12], cosmic expansion (Pantheon+ SNe Ia and DESI BAO, Pillar 1) [14], and early-universe structure growth (JWST, Paper 2.1) [18], the empirical success of Chronodynamic Relativity in wide binary stars establishes a single, unified, parameter-free gravitational paradigm spanning stellar, galactic, and cosmological scales.

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## DATA AND CODE AVAILABILITY

The complete source code, physics engine, Monte Carlo orbital integrator, and data processing scripts supporting this study are openly available in the companion repository at `https://github.com/joshreve/Chronodynamic-Relativity`. A permanent, immutable version of record for the software framework is archived on Zenodo under Concept DOI: 10.5281/zenodo.22697449. This manuscript and its associated reproduction assets are permanently archived on Zenodo under DOI: 10.5281/zenodo.23149642.

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